

Jack Pham

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EDUCATION

University of California, Berkeley

Bachelor of Science in Electrical Engineering and Computer Sciences, GPA: 3.54

Berkeley, CA
August 2025 - May 2028 (Expected)

Cypress College

Associate Degree for Transfer in Mathematics and Physics, GPA: 3.94

Cypress, CA
February 2023 - May 2025

TECHNICAL SKILLS

Programming & Data Analysis: Python, C++, Rust, SQL, MATLAB, NumPy, SciPy, scikit-learn, XGBoost

RF & Semiconductor Test: Network Analyzer & Spectrum Analyzer RF Test, GaAs HBT & SAW Device Characterization, STDF Data Pipelines, SPC / Cpk Yield Analysis

Engineering Tools: JMP, Exensio, Git, Linux, Unix, Bash, Cursor, KiCad, LTspice, AutoCAD, HPC Systems

WORK EXPERIENCE

Skyworks Solutions

Product Engineer Co-Op

Irvine, CA

June 2026 - Present

- Investigate product and manufacturing issues across semiconductor die, wafer, and module test data, applying Cpk, SPC, and ANOVA statistical yield-analysis methods to isolate yield trends, process shifts, and root causes of performance failures on high-volume RF front-end products.
- Characterize RF chip performance on GaAs HBT and SAW power amplifier and low-noise amplifier devices using network and spectrum analyzers, measuring insertion loss, gain, and harmonic distortion during Final Test.
- Build a traceable Parameter Review workbook in Python (pandas, openpyxl) merging raw FT-SPEC files, Test Name Standards, and the STC engineering database across ~44,000 test rows and 100 Test Types, applying source-priority judgment to flag spec conflicts for owner review instead of guessing.
- Build and maintain the ARE limit-evaluation tool and STDF-to-CSV pipelines in Python, C++, and Rust, accelerating processing of 3,000+ rows of production test data across seven-plus vendor formats while collaborating with 8 engineers across design, test, manufacturing, and quality using JMP, Exensio, and Cursor.

North Orange County Community College District

Physics Tutor

Cypress, CA

February 2025 - July 2025

- Developed Python and Jupyter simulations to visualize electromagnetism, relativity, and thermodynamics concepts across 40+ tutoring hours, contributing to a 20% average improvement in student exam performance, and supported physics lab sessions on circuits and PCB concepts using multimeters and other instrumentation.

RESEARCH EXPERIENCE

Texas Advanced Computing Center (TACC), UT Austin

Summer Researcher

Austin, TX

June 2024 - August 2024

- Formulated multi-robot task allocation as a linear program minimizing the maximum task-load discrepancy across 5 heterogeneous agents (Scout, Medic, Engineer, Carrier, FireBot) in a simulated disaster-response grid, combining it with a Fairness-Enhanced MADDPG reward function to improve task-distribution efficiency by 70% over the baseline.
- Modeled agent coordination as a Markov Decision Process and implemented Conflict-Based Search (CBS) pathfinding integrated with the LP task allocator to generate collision-free, cost-minimized paths around obstacles and fire tasks.
- Ran large-scale optimization experiments on TACC's Linux/Unix HPC cluster, submitting model runs and managing data transfer via the command line; the LP-based approach reduced average solve time by 45% with lower variance than the non-LP baseline.

PROJECTS

AccessiSound: Sound Assistive Device | [GitHub](#) - Embedded C++, TensorFlow Lite Micro, ESP32-S3

Spring 2026

- Engineered a \$17 offline assistive device on ESP32-S3 that extracts MFCC features (Hamming windowing, Mel filterbank) to detect doorbells, fire alarms, and other household sounds for visually impaired users.
- Trained a compact CNN on the ESC-50 dataset and deployed it via int8 quantization to a ~45 kB TensorFlow Lite Micro model, reaching ~80% accuracy across 5 sound classes at ~50ms inference latency.

Methane Emission Monitoring Platform, Fung Fellowship | [Fung Institute](#)

UC Berkeley, Spring 2026

- Co-designed a \$15 household methane sensor from off-the-shelf components and conducted 12 stakeholder interviews with Climate Action California to close a data gap in residential-level methane tracking.
- Worked with a 4-person cross-disciplinary team to fold stakeholder feedback into the sensor design and submitted findings to the California Air Resources Board to inform statewide methane mitigation policy.

HONORS & ACTIVITIES

Underwater Robotics Team, UC Berkeley

Electrical Subteam Member

Berkeley, CA

February 2026 - Present

- Investigate power dissipation and thermal behavior of NVIDIA Jetson development modules used in autonomous underwater robots, and support electrical subsystem integration including power regulation, embedded hardware interfacing, and system wiring.
- Assist with debugging and testing of onboard computing systems connected to 16 sensors and actuators, and collaborate with the mechanical subteam to design and prepare 3D-printed fixtures and enclosures optimized for space, waterproofing, weight, and thermal performance.
- PEARC26 Student Program Participant, TACC. UC Berkeley Fung Fellow, Spring 2026. Presenter, SCCUR and STARS Celebration Conference, 2024.